

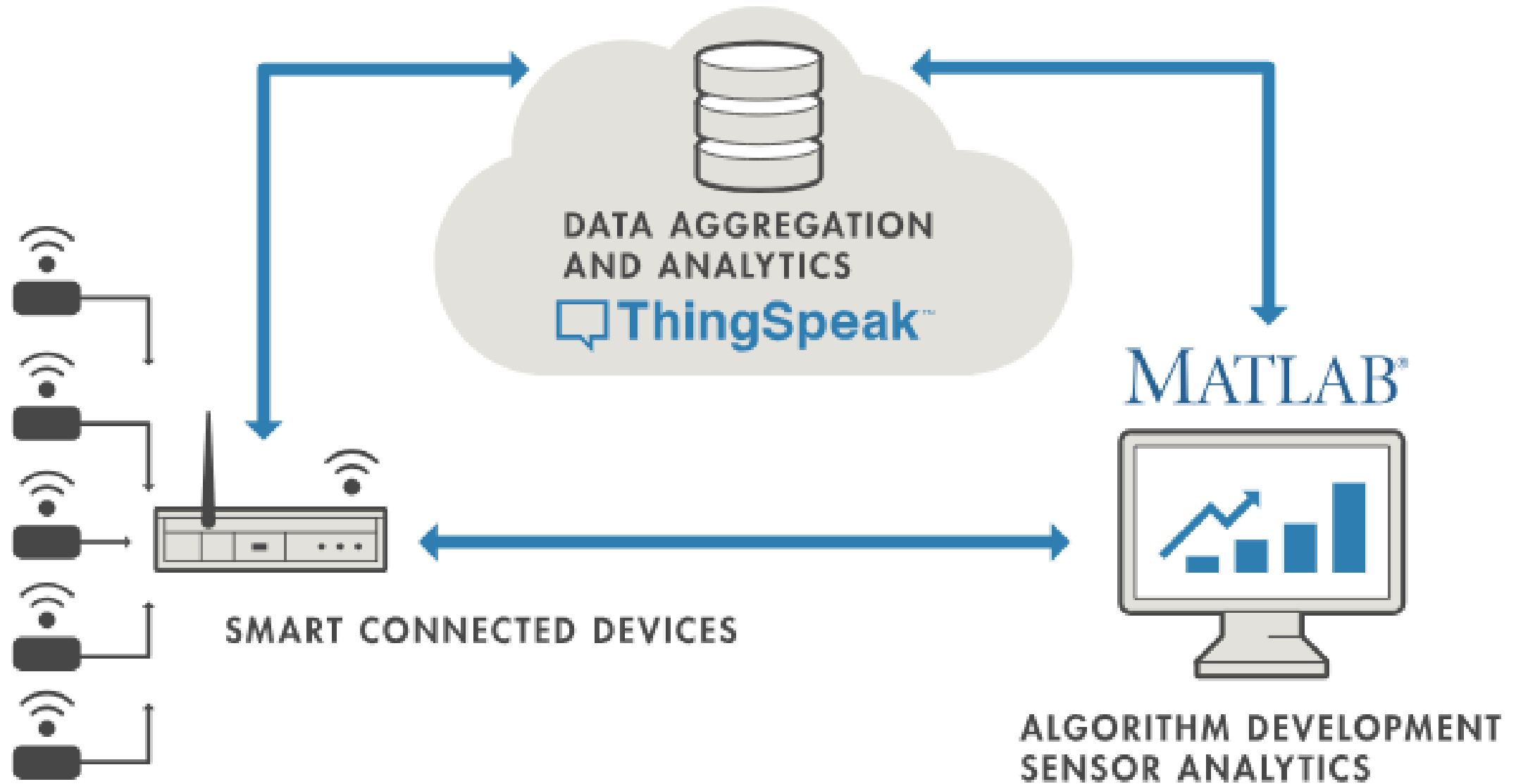


Naan Mudhalvan

Industrial IoT and Industry 4.0

Tektork Private Limited, Coimbatore

Cloud Platform - ThingSpeak



ThingSpeak is a cloud-based IoT analytics platform that enables live data collection, visualization, analysis, and automated actions, streamlining the development of sensor-based connected applications without requiring users to build or host their own web infrastructure.

It acts as the bridge between resource-constrained hardware and sophisticated cloud-based data processing.

Core Theory and Architecture

- ThingSpeak operates by letting IoT devices (sensors, actuators, gateways, edge nodes) send data to the platform via secure web protocols (HTTP REST API, MQTT), commonly using libraries for platforms like Arduino, ESP8266, ESP32, Particle, and Raspberry Pi.
- Each data flow is organized in “channels,” which are virtual containers storing up to 8 numerical fields and optional location data (latitude, longitude, elevation) per timestamped entry.

Features and Data Operations

- **Live Data Aggregation:**
 - Devices continuously post real-time sensor or control data to dedicated ThingSpeak channels.
- **Visualization:**
 - Built-in charting, gauges, historical and geographic maps for live/historic data monitoring; customizable with widgets and MATLAB visualizations.
- **Analysis & Automation:**
 - Scheduled executions of MATLAB algorithms for data analysis, event detection, predictive functions, and DSP tasks.

- **Third-party Actions:**
 - Integrations with web services (e.g., Twitter, Twilio), allowing triggers and notifications based on data conditions.
- **Open Source and Extensible:**
 - ThingSpeak encourages custom plugins and integrations for specific IoT use cases, supporting innovation and collaborative development.

Core Components of ThingSpeak

Component	Description
Channel	A virtual container where your data is stored. Each IoT project usually has one channel.
Field	Each channel has up to 8 data fields. Each field stores a specific type of data (e.g., temperature, humidity, LDR).
Channel ID	A unique number assigned to your channel — used to send and read data.
Write API Key	A secret key used to send data to the channel.
Read API Key	A key used to read data from the channel (for dashboards, apps, etc.).
Feeds	Each data entry (timestamp + field values) is called a feed.
Public/Private Channels	Channels can be made public (for anyone to view) or private (only with API key).

Workflow: Creating and Using Channels

- **Sign up on ThingSpeak and create a channel:**
 - Assign a name, description, and define field names based on what your device will send (e.g., "Temperature", "Humidity").
- **Configure your device/software to POST data:**
 - Use the channel's Write API key and construct the update request (e.g., via
- HTTP `api.thingspeak.com/update?api_key=YOUR_KEY&field1=value1`). Send single data points or batches.
- **View data instantly:**
 - ThingSpeak displays live charts and visualizations for all fields; historical data can be explored and exported for analysis.

Workflow: Creating and Using Channels Cond..

- **Read data from channels:**
 - Devices or users can fetch past/historic data using the Read API key.
- **MATLAB analytics in cloud:**
 - Run MATLAB scripts directly on your ThingSpeak data for advanced processing (filtering, event detection, forecasting, etc.).

Data Visualization and Analysis

- ThingSpeak offers built-in charts, gauges, and geographic maps. These provide instant visual feedback and easy monitoring of trends and events.
- You can customize visualizations using MATLAB Visualizations App, or embed widgets in external sites/tools.
- Automated data actions and triggers: React Apps and TimeControl Apps let you set up automatic responses.
- For example, send alerts, log events, or change device operation based on condition rules.

- **REST API:**
 - Simple HTTP requests to send (update) or receive (read) data. Supports query parameters for field values and location.
- **MQTT API:**
 - For devices needing publish/subscribe message-based communication.
- **Cloud integrations:**
 - Forward data from LoRaWAN, Particle, The Things Network, and other platforms using bridge tools.

Security and Access

- Every channel has independent Write and Read API keys, controlling who can send or retrieve data.
- Channels can be set public/private; public channels let others view or analyze data for collaborative projects.

Limitation of ThingSpeak

Limit	Free Plan
Minimum update interval	15 seconds
Channels per account	4
Fields per channel	8
Data storage	Unlimited, but slow query after 3 million records
MATLAB execution	20,000 characters per script

Advantages

- No infrastructure to maintain: Thingspeak abstracts backend complexity.
- Powerful MATLAB analytics built-in without additional environment setup.
- Flexibility to connect diverse IoT hardware and protocols easily.
- Facilitates educational, prototyping, and industrial IoT applications.

Example Use Cases

- Weather stations
- Smart agriculture
- Home automation
- Energy monitoring
- Health tracking

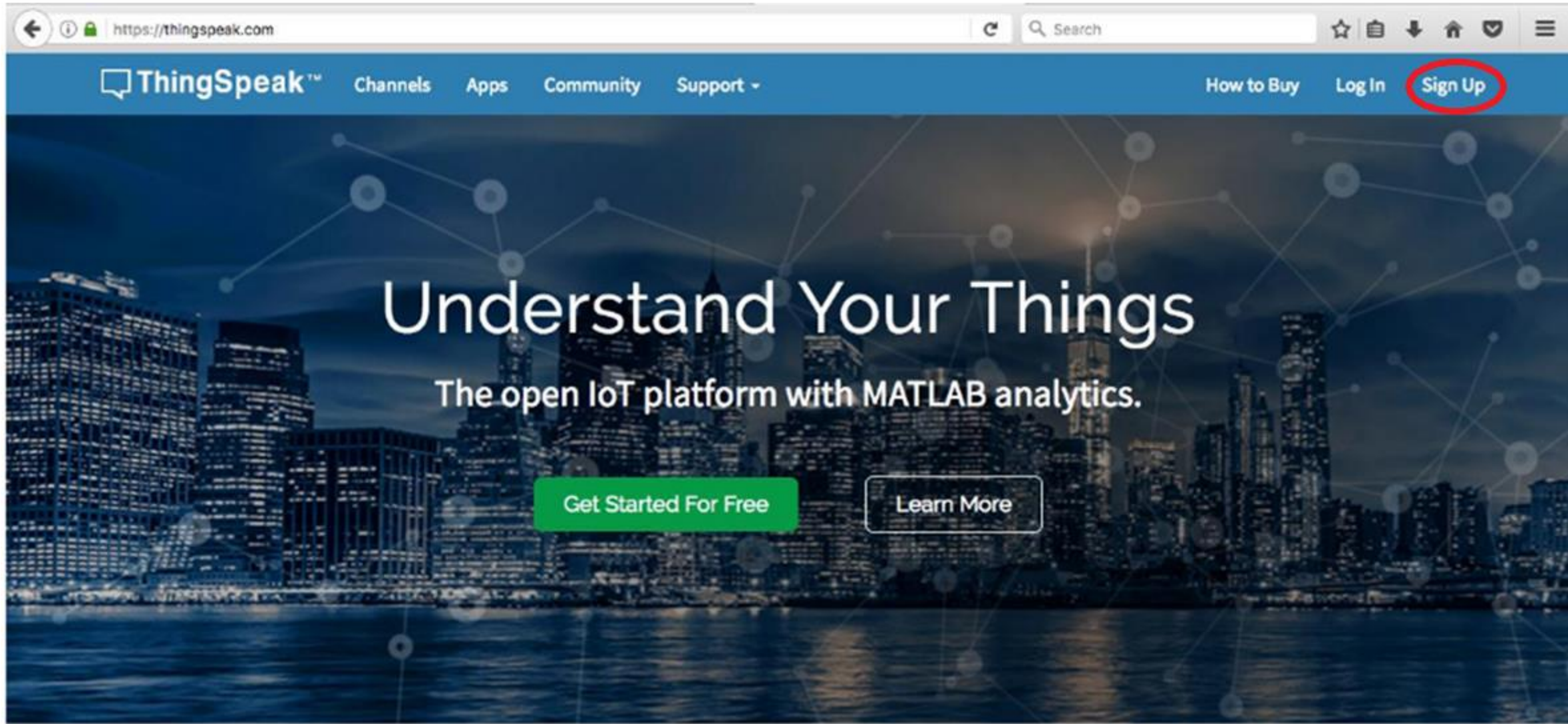
ThingSpeak

- Go to the ThingSpeak website and click the "**Get Started for Free**" button to create a new account.
- This account is linked to your **MathWorks** account, so if you already have one, simply log in using that account instead of creating a new one.

- Open your web browser.
- Go to <https://www.thingspeak.com>.
- Your browser should display the page shown here:



- Click the **Sign Up** button located in the upper-right corner of the page.



- Fill in the required fields: email address, username, password, country, first name, and last name, then click the **Continue** button.

Sign up for ThingSpeak

The ThingSpeak service is operated by MathWorks. In order to sign up for ThingSpeak, you must create a new MathWorks Account or log in to your MathWorks Account.

Create MathWorks Account

Email Address

United States

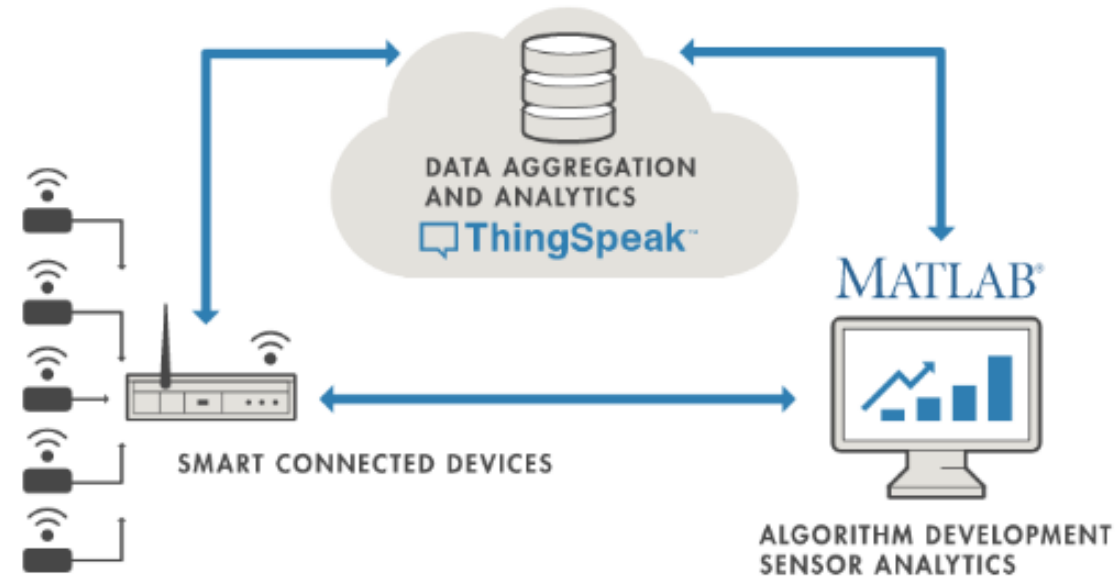
First Name

Last Name

By clicking continue, you agree to our [privacy policy](#)

Cancel

Continue



- After clicking the **Continue** button, your screen will look like this.


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Verify Your MathWorks Account

To finish creating your account, complete the following steps:

- Go to your inbox for

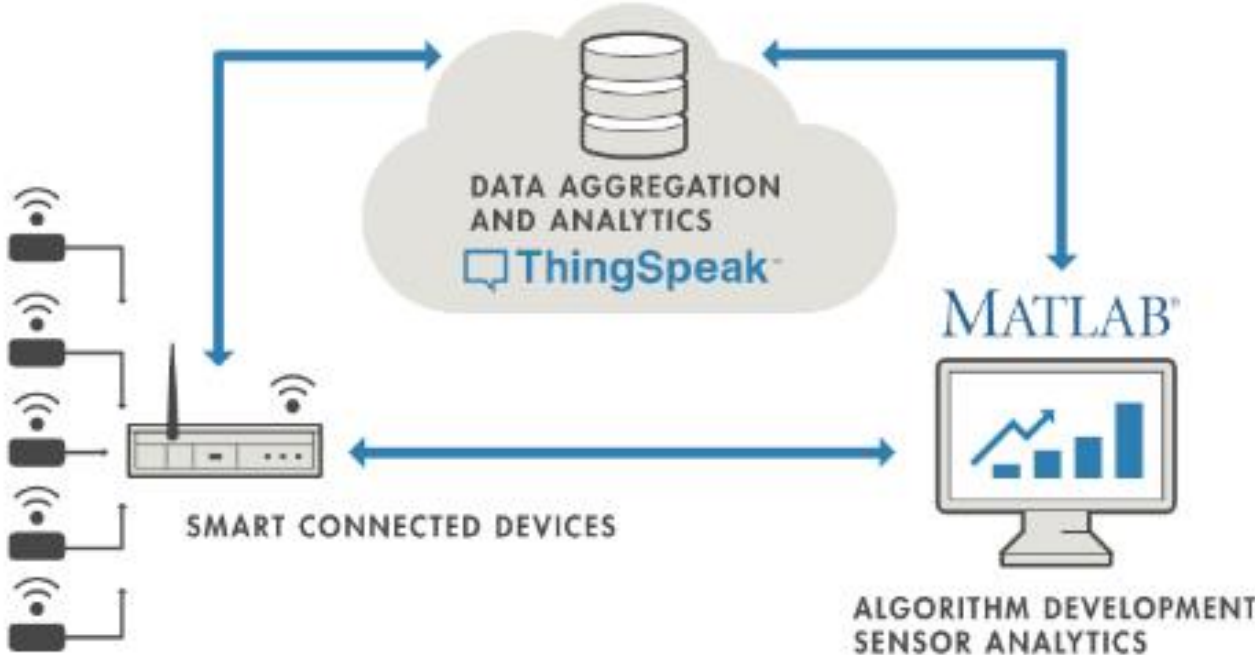
Your email Id displayed here

- Click the link in the email we sent you.

Once you've done this, click Continue.

Didn't get the email?

- Check your spam folder.
- [Send me the email again.](#)
- Contact Customer Support if you still do not have the email



- Open a new tab and check your email inbox for a message from "service" with the subject "Verify email address." Open this email and proceed to **Next Step**

***If the email is not in your inbox, check your spam folder. If it's not there, choose the "Send me the email again" option in the window with ThingSpeak (see image below).**


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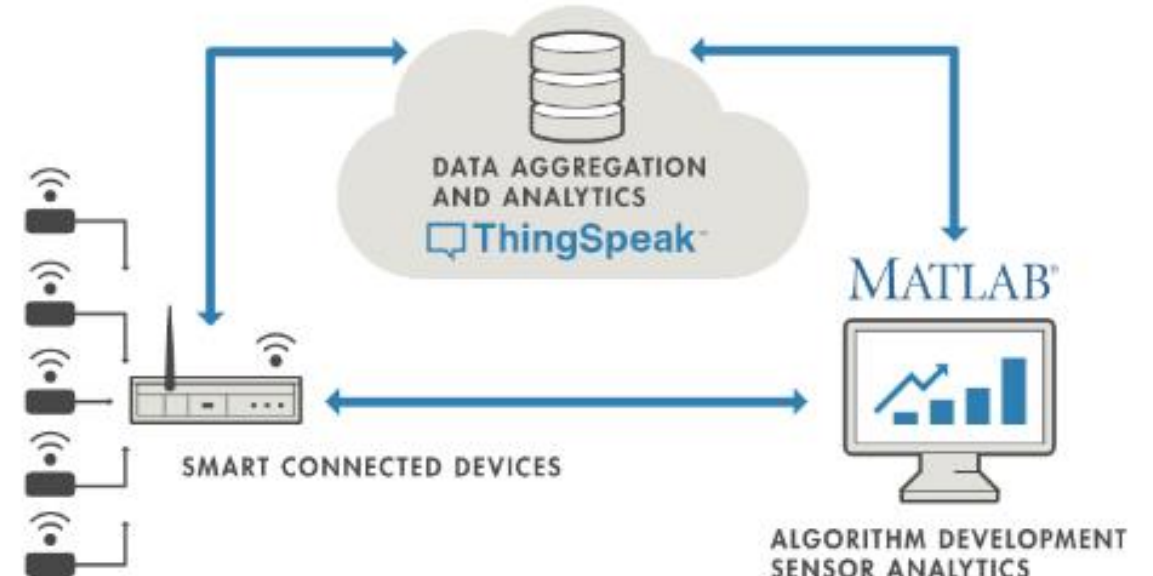
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- The message should look like the one shown below. Click “**Verify Your Email.**”

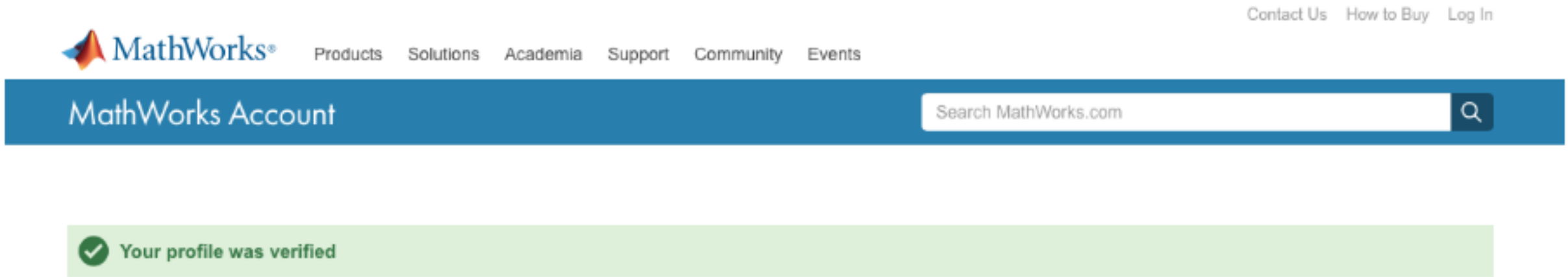
Thank you for registering with MathWorks!

To complete the registration process, verify your email address by clicking this link:



Sincerely,
MathWorks Customer Service Team

- A confirmation page will appear. Notice that it opens a new tab for **MathWorks Inc.**, the parent company of ThingSpeak.



- Go back to www.thingspeak.com and click **Log In**.



- Enter your email address and password, then click “Sign In.”

 **ThingSpeak™**

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Log in to ThingSpeak

Email Address

Password

[Forgot your password?](#)

[Sign In](#)

- A confirmation page will appear. Click **OK** to proceed.

Signed in successfully.

X

Sign-up successful

Congratulations, you have successfully linked your MathWorks account to ThingSpeak. Use the following email ID and its associated MathWorks account password on all subsequent logins to ThingSpeak.

Email ID: **wagne1at@gmail.com**

Welcome to ThingSpeak!

OK

- Read and agree to the **Terms of Use**.

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ThingSpeak Terms of Use

ThingSpeak Terms of Use have changed. We require that you agree to the [Terms of Use](#) and [Privacy Policy](#) before continuing.

[Agree to Terms](#)[Decline and Sign Out](#)

- Congratulations! If your screen looks like the one below, you have successfully registered for **ThingSpeak**.

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My Channels

New Channel

Help

Collect data in a ThingSpeak channel from a device, from another channel, or from the web.

Click **New Channel** to create a new ThingSpeak channel.

Click on the column headers of the table to sort by the entries in that column.

Learn to [create channels](#), explore and transform data.

Learn more about [ThingSpeak Channels](#).

Examples

- [Arduino](#)
- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)
- [Netduino Plus](#)

Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

Upgrade

- Click New Channel



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Examples

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- [Arduino MKR1000](#)
- [ESP8266](#)
- [Raspberry Pi](#)
- [Netduino Plus](#)

Upgrade

Need to send more data faster?

Need to use ThingSpeak for a commercial project?

Upgrade

- Create a single channel to collect the raw values from the **potentiometer**.

New Channel

Name	Potentiometer Value (ADC)	
Description		
Field 1	Potvalue	☒
Field 2		☐
Field 3		☐

Help

Channels store all the data that a ThingSpeak application collects. Each channel includes eight fields that can hold any type of data, plus three fields for location data and one for status data. Once you collect data in a channel, you can use ThingSpeak apps to analyze and visualize it.

Channel Settings

- **Channel Name:** Enter a unique name for the ThingSpeak channel.
- **Description:** Enter a description of the ThingSpeak channel.
- **Field#:** Check the box to enable the field, and enter a field name. Each ThingSpeak channel can have up to 8 fields.

- Scroll to the bottom and click “Save Channel.”

ThingSpeak™

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Sign Out

Show Location

☐

Latitude

Longitude

Show Video

☐

☒ YouTube

☐ Vimeo

Video ID

Show Status

☐

Save Channel

- The screen should appear as shown below. Then, click the “API Keys” button.


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Primary Channel

Channel ID:
Author:
Access: Private

This channel will collect data from the ultrasonic sensor.

[Private View](#)
[Public View](#)
[Channel Settings](#)
[Sharing](#)
[API Keys](#)
[Data Import / Export](#)

[+ Add Visualizations](#)
[Data Export](#)
[MATLAB Analysis](#)
[MATLAB Visualization](#)

Channel Stats

Created: [less than a minute ago](#)
Updated: [less than a minute ago](#)
Entries: 0

Field 1 Chart

Potentiometer Value (ADC)

- This is where the **API Keys** and the **Channel ID** for this channel are located. We will need all three of these for our code: the **Channel ID**, the **Write API Key**, and the **Read API Key**.


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Primary Channel

Channel ID: **306964**
Author:
Access: Private

This will process data from the ultrasonic sensor.

[Private View](#)
[Public View](#)
[Channel Settings](#)
[Sharing](#)
[API Keys](#)
[Data Import / Export](#)

Write API Key

Key

FYLP4EYMV9ICFMFQ

Generate New Write API Key

Read API Keys

Key

3S6H9W7ZNN2CWUTY

Help

API keys enable you to write data to a channel or read data from a private channel. API keys are auto-generated when you create a new channel.

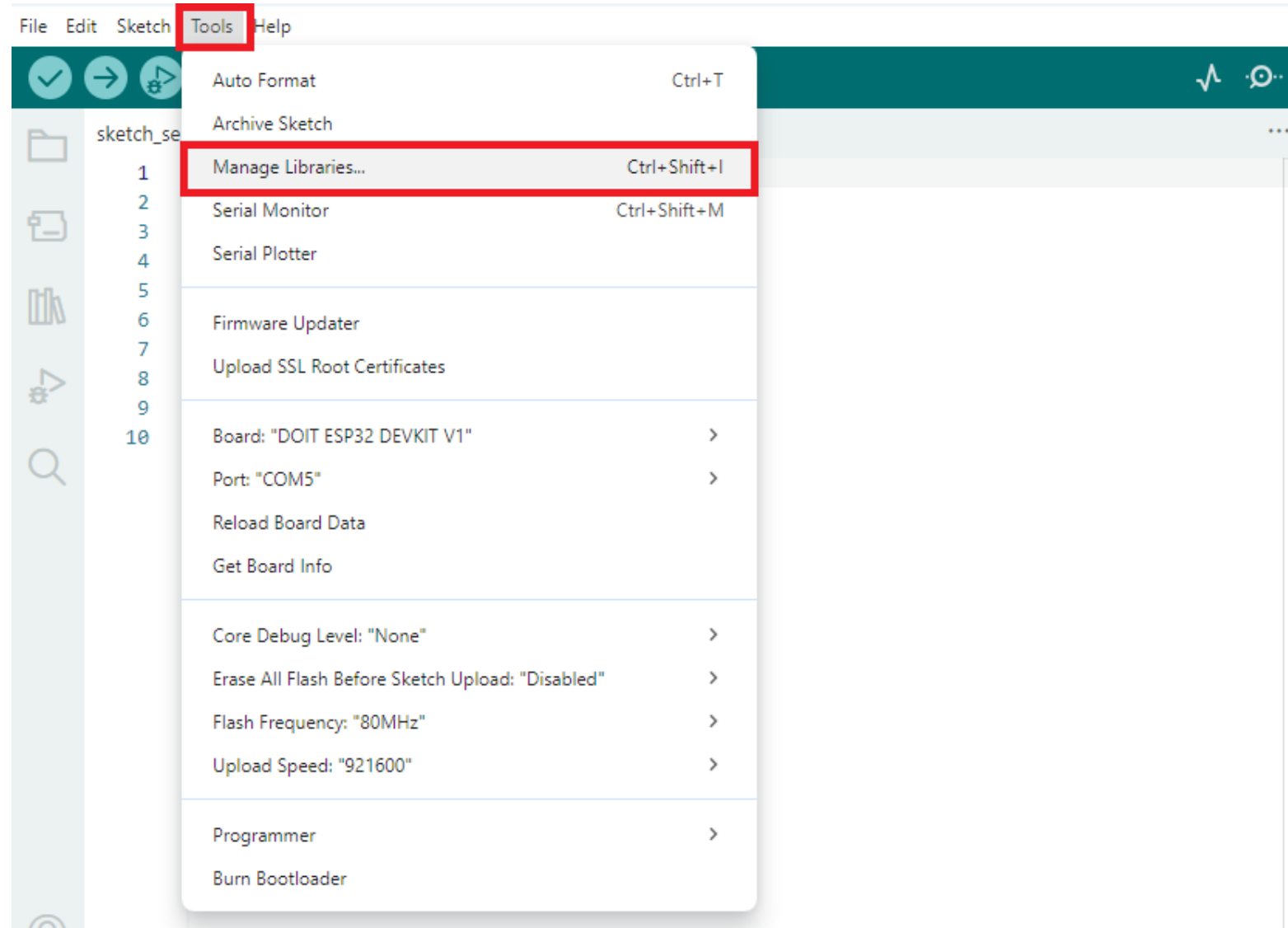
API Keys Settings

- Write API Key:** Use this key to write data to a channel. If you feel your key has been compromised, click **Generate New Write API Key**.
- Read API Keys:** Use this key to allow other people to view your private channel feeds and charts. Click **Generate New Read API Key** to generate an additional read key for the channel.
- Note:** Use this field to enter information about channel read keys. For example, add notes to keep track of users with access to your channel.

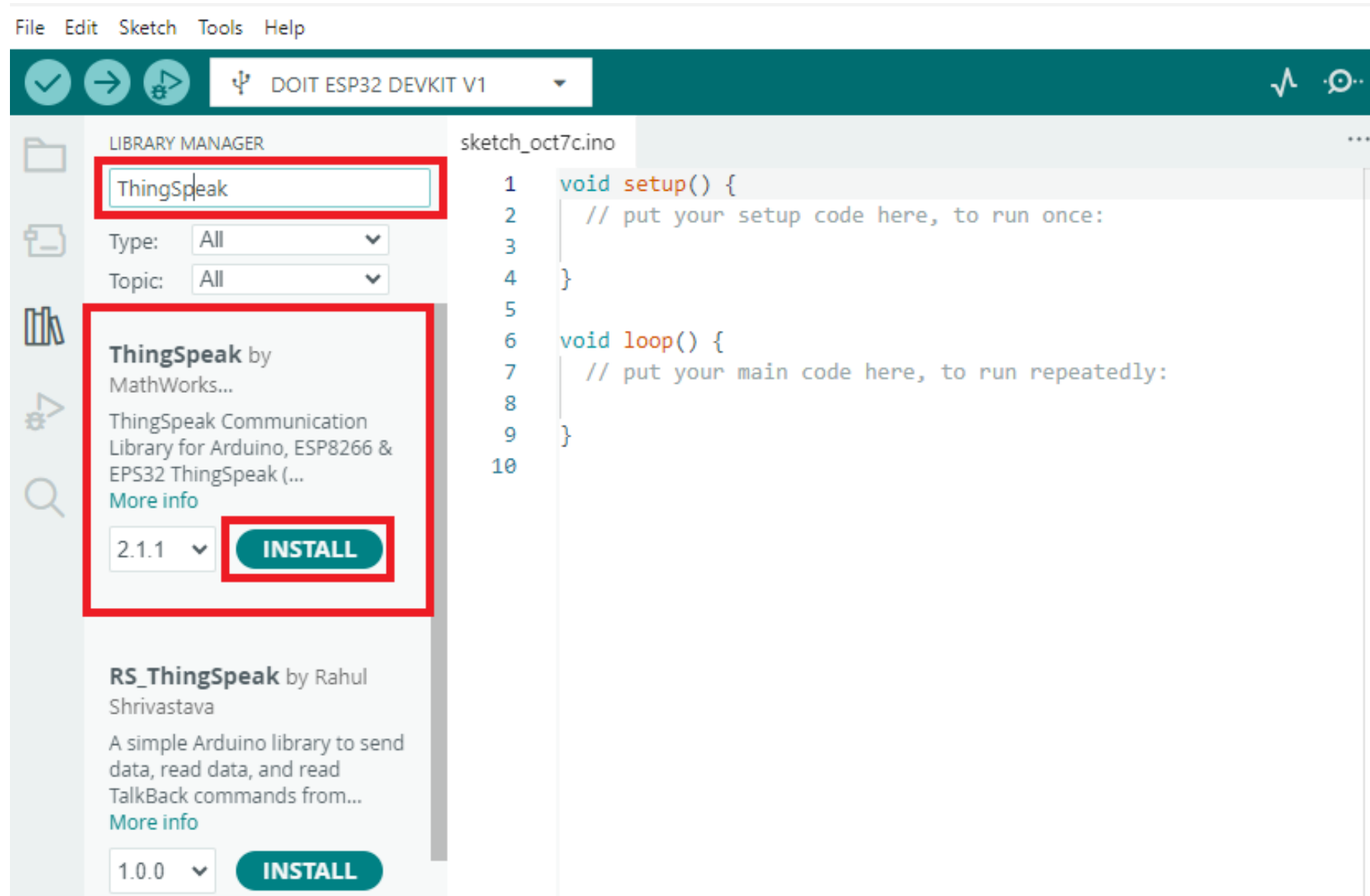
API Requests

ThingSpeak - Library installation

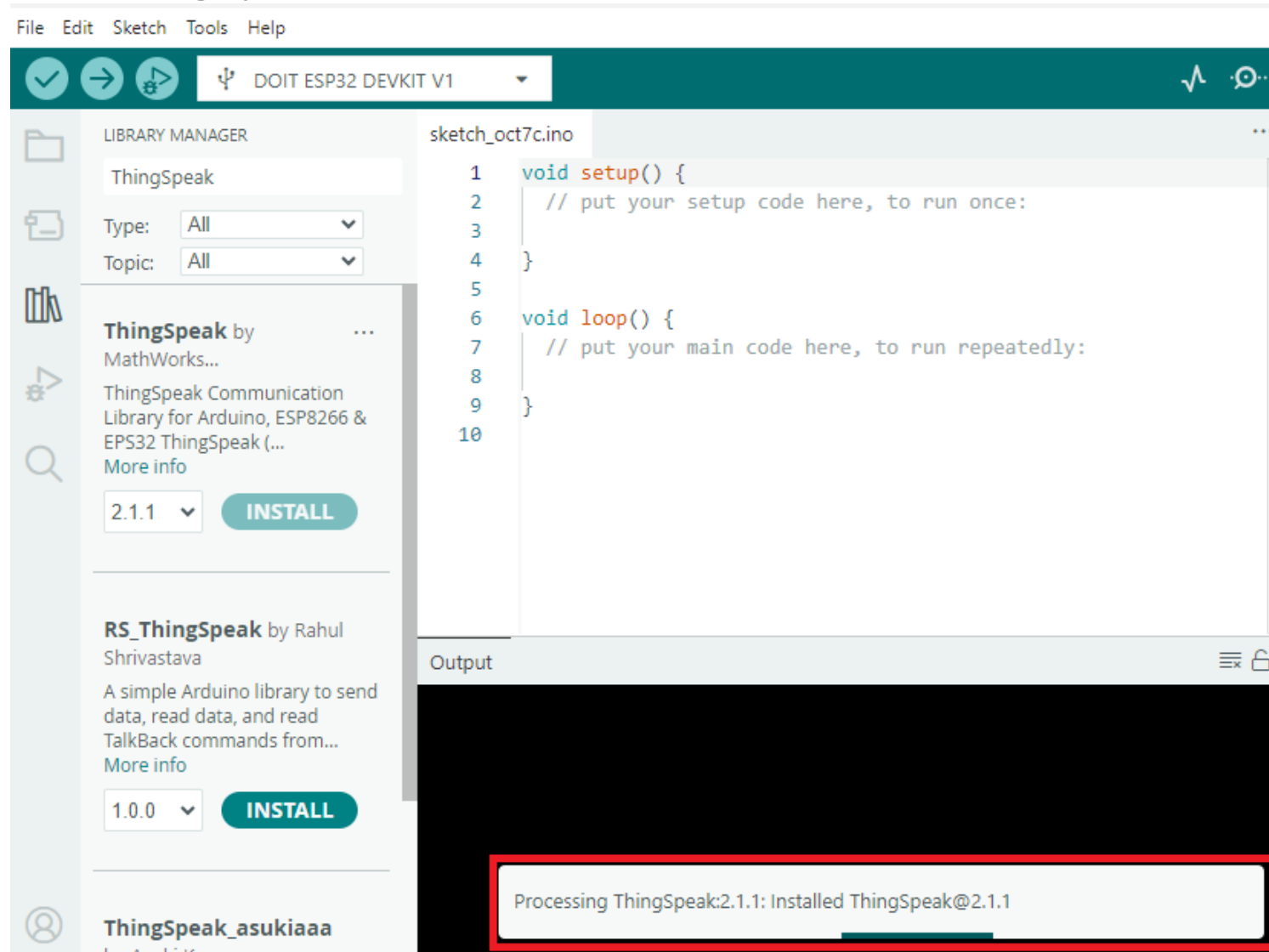
- In the Arduino IDE, go to the Tools menu, then select Manage Libraries... to open the Library Manager.



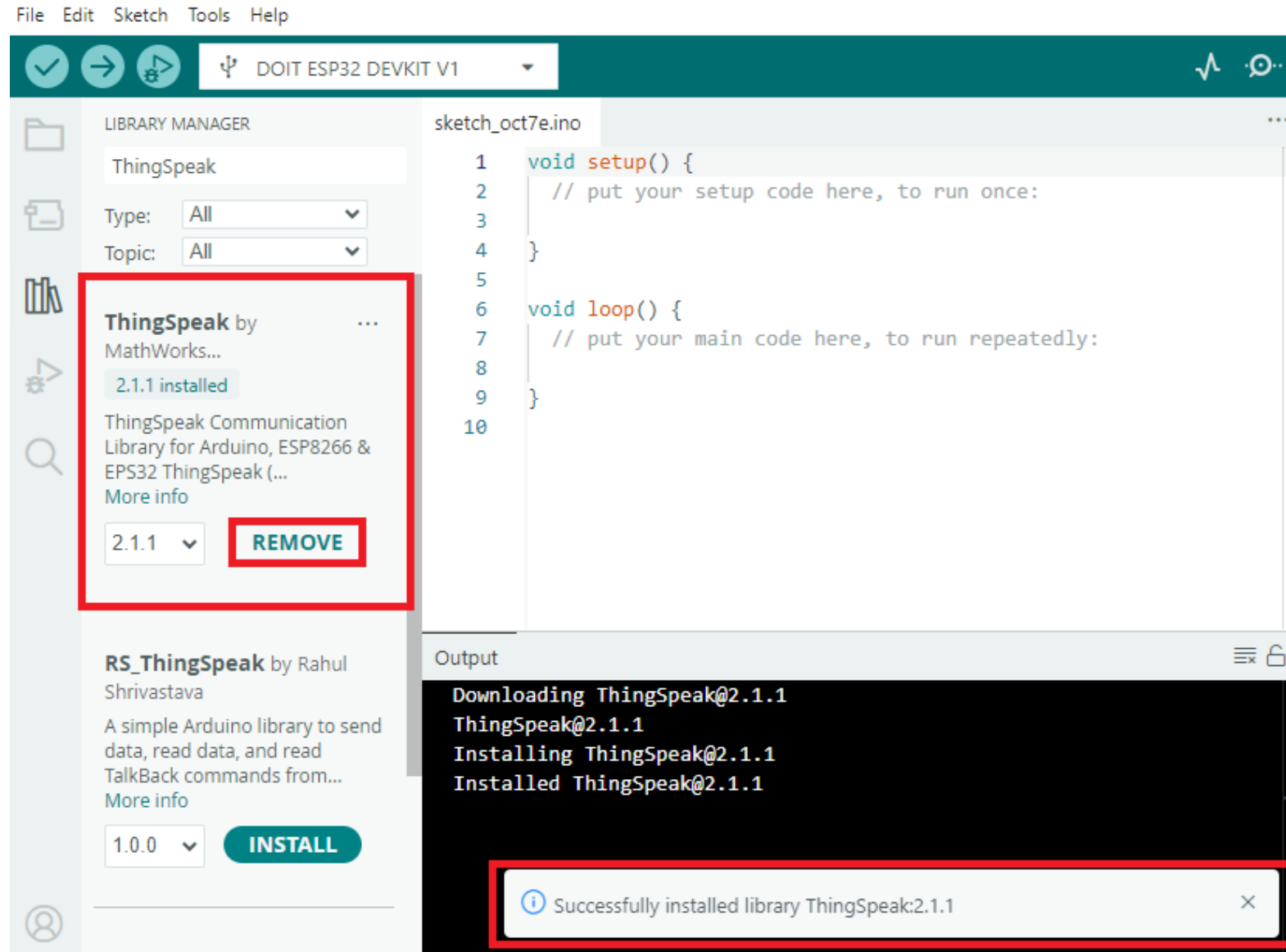
- In the Library Manager search box, type **ThingSpeak**, then select the **ThingSpeak** library and click **Install**.



- The installation of the ThingSpeak library is in progress. Wait until the installation is complete. Once it finishes, you can include the library in your Arduino sketch to send sensor data to the ThingSpeak cloud.



- After the installation is complete, the Install button will change to Remove, and the output window will display the installation details.



ThingSpeak - Potentiometer Code

```
#include <WiFi.h>           // Include Wi-Fi library for ESP32
#include <ThingSpeak.h>      // Include ThingSpeak library for IoT data communication

// Wi-Fi credentials - replace with your network's SSID and password
const char* ssid = "Your SSID";
const char* password = "Your Password";

WiFiClient client;          // Create a Wi-Fi client object for network communication

// ThingSpeak channel configuration
unsigned long myChannelNumber = Your Channel Number;    // Your ThingSpeak Channel ID
(replace this)
const char* myWriteAPIKey = "Your Write API Key"; // Your ThingSpeak Write API Key (replace
this)

// Potentiometer connection pin on ESP32
const int potPin = 36;      // GPIO 3 analog input connected to potentiometer output

// Variables for timing data uploads to ThingSpeak
unsigned long lastTime = 0;    // Stores the timestamp of the last upload
unsigned long timerDelay = 20000; // Delay between uploads (20 seconds, free ThingSpeak limit)
```

```
void setup()
{
    Serial.begin(115200);    // Start serial communication at 115200 baud for debugging

    // Connect to Wi-Fi as station mode
    WiFi.mode(WIFI_STA);
    WiFi.begin(ssid, password);
    Serial.print("Connecting to WiFi ..");
    // Wait in a loop until connection is established
    while (WiFi.status() != WL_CONNECTED) {
        Serial.print('.');    // Print dots as connection attempts
        delay(1000);          // Wait 1 second before retrying
    }

    // Connection successful - print confirmation and network details
    Serial.println("\n Connected successfully!");
    Serial.print(" SSID: ");
    Serial.println(WiFi.SSID());    // Print connected Wi-Fi SSID
    Serial.print(" IP Address: ");
    Serial.println(WiFi.localIP());    // Print assigned ESP32 IP address
    ThingSpeak.begin(client);    // Initialize ThingSpeak using the Wi-Fi client
}
```

```
void loop() {  
    // Execute this block only after timerDelay time has elapsed  
    if ((millis() - lastTime) > timerDelay) {  
  
        // Check if Wi-Fi is still connected  
        if (WiFi.status() != WL_CONNECTED) {  
            Serial.print("Attempting to reconnect to WiFi");  
            WiFi.begin(ssid, password);  
            // Reconnect until successful  
            while (WiFi.status() != WL_CONNECTED) {  
                Serial.print('.');  
                delay(5000);  
            }  
            Serial.println("\nReconnected.");  
  
            // Print Wi-Fi connection details after reconnecting  
            Serial.print(" SSID: ");  
            Serial.println(WiFi.SSID());  
            Serial.print(" IP Address: ");  
            Serial.println(WiFi.localIP());  
        }  
    }  
}
```

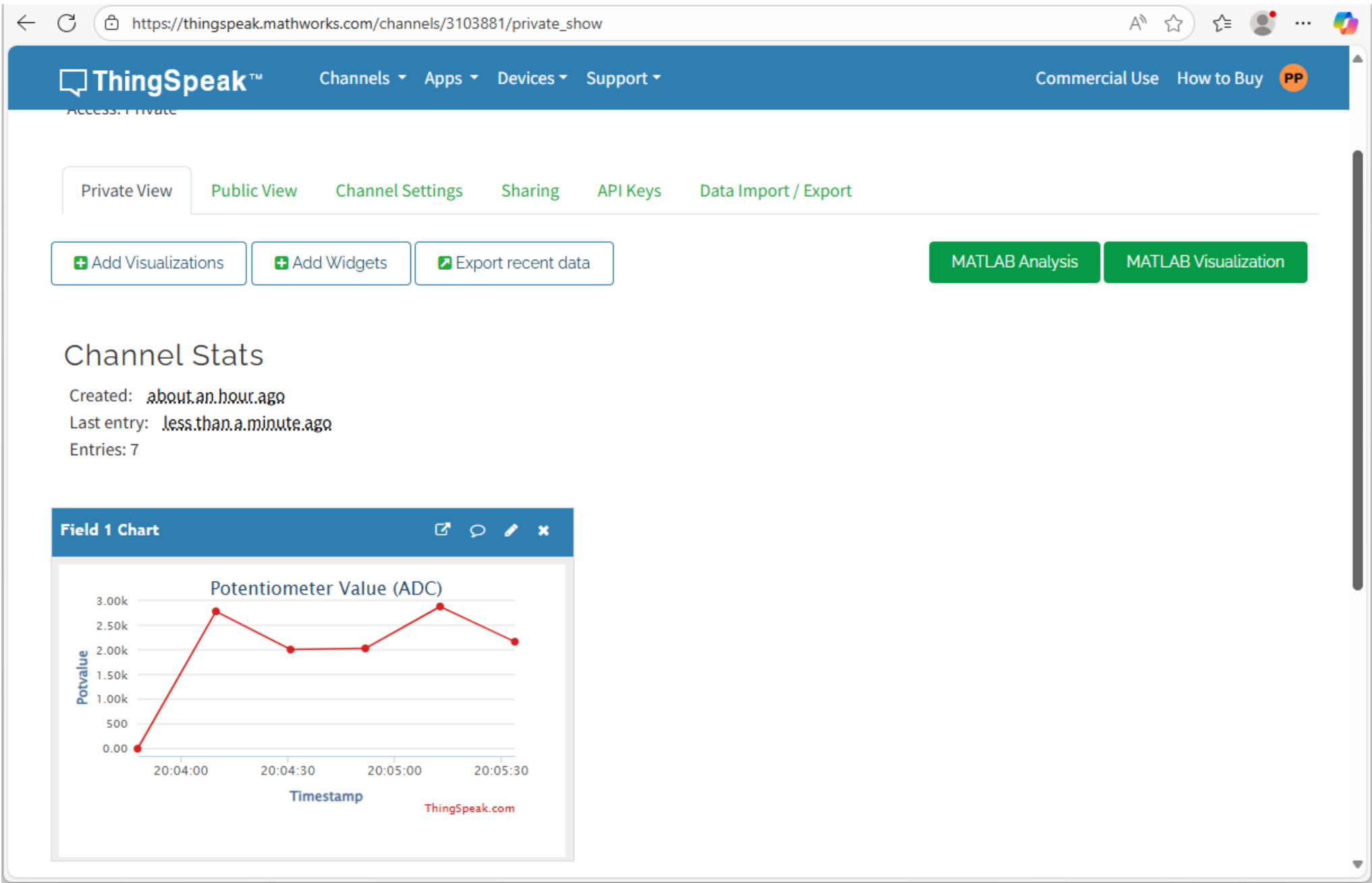


```
// Read analog potentiometer value (range from 0 to 4095)
int potValue = analogRead(potPin);
Serial.println("Potentiometer Value: " + String(potValue)); // Print value to serial
monitor

// Write potentiometer value to ThingSpeak channel Field 1
int x = ThingSpeak.writeField(myChannelNumber, 1, potValue, myWriteAPIKey);

// Check response of ThingSpeak update attempt
if (x == 200) {
    Serial.println("Channel update successful."); // Success message
} else {
    Serial.println("Problem updating channel. HTTP error code " + String(x)); // Error
message
}

lastTime = millis(); // Update last upload time to current time
}
}
```



Thank You